

Grade 10 Fundamentals – Mathematical Notation

Contents

1	Worked Solution Examples	1
2	Conceptual Checkpoints and Summary	7

Evaluated criteria

C0 Interpret and use basic mathematical notation for quantities, patterns, variables, ellipses, and indexed collections.

1 Worked Solution Examples

Problems:

- 1.1. Written Lists and Patterns (C0)
 - 1.2. Ellipsis Notation (C0)
 - 1.3. From Names to Variables (C0)
 - 1.4. Subscripts and Collections (C0)
 - 1.5. Generalizing the Size of a Collection: n (C0)
 - 1.6. Generalizing a Position: i (C0)
 - 1.7. Using Two Indices: i and j (C0)
 - 1.8. Combining Indices and Ellipses (C0)
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1.1 Written Lists and Patterns (C0)

We often begin by writing every number in a list. Look at these examples:

1, 2, 3, 4, 5 2, 4, 6, 8, 10 10, 20, 30, 40, 50 1, 2, 4, 8, 16.

Each list follows a pattern. A **pattern** is a rule that tells us how the entries are related.

Questions/Tasks.

C0 Write the next three entries of 1, 2, 3, 4, 5.

C0 Write the next three entries of 2, 4, 6, 8, 10.

C0 Write the next three entries of 10, 20, 30, 40, 50.

C0 - Continue the list 1, 2, 3, 4, 5.	C0 - Continue the list 2, 4, 6, 8, 10.	C0 - Continue the list 10, 20, 30, 40, 50.
The numbers increase by 1 each time. The next three entries are	The numbers increase by 2 each time. The next three entries are	The numbers increase by 10 each time. The next three entries are
6, 7, 8.	12, 14, 16.	60, 70, 80.

When a pattern is clear, mathematics can save space by not writing every entry. This leads to the ellipsis notation ...



1.2 Ellipsis Notation (C0)

The symbol

...

is called an **ellipsis**. It means that some entries have been left out because the surrounding entries make the intended pattern clear.

For example,

$$1, 2, 3, \dots, 10 = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.$$

The ellipsis does **not** always mean “keep adding 1.” The pattern comes from the numbers around it.

2, 4, 6, ..., 30	5, 10, 15, ..., 75	30, 28, 26, ..., 2
means that the even numbers continue up to 30.	means that multiples of 5 continue up to 75.	means that the numbers decrease by 2 until 2.
2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30.	5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75.	30, 28, 26, 24, 22, 20, 18, 16, 14, 12, 10, 8, 6, 4, 2.

Questions/Tasks.

C0 Write 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 using ...

C0 Write 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36 using ...

C0 Write 50, 45, 40, 35, 30, 25, 20, 15, 10, 5 using ...

C0 - Write the list using ...	C0 - Write the list using ...	C0 - Write the list using ...
The numbers increase by 1:	The numbers increase by 3:	The numbers decrease by 5:
1, 2, 3, ..., 12.	3, 6, 9, ..., 36.	50, 45, 40, ..., 5.



Checkpoint: Does ... define the pattern? No. The dots only show that entries were omitted; the surrounding entries must make the pattern clear. For example,

$$2, 4, 6, \dots, 20 \quad 5, 10, 15, \dots, 50 \quad 30, 27, 24, \dots, 3$$

are clear from their first entries. But

$$2, 4, \dots, 16$$

is ambiguous: it could continue by adding 2 or by multiplying by 2. Use ... only when the intended pattern is sufficiently clear.



1.3 From Names to Variables (C0)

A quantity can have a long name. A variable gives it a short symbolic name:

$$\text{Savings} = 20 \quad \longrightarrow \quad S = 20.$$

The letter does not have a fixed meaning. We define it and then use it consistently.

$$P = \text{price of one sandwich}, \quad Q = \text{number of sandwiches sold}, \quad R = \text{money collected}.$$

Then

$$R = P \times Q.$$

A **variable** is a symbol used to represent a quantity. Its value can change while its meaning stays the same:

$$Q = 2, \quad Q = 5, \quad Q = 10.$$

Questions/Tasks.

C0 For the quantity “number of customers,” suggest a convenient single-letter variable and state what it represents.

C0 If C = number of customers, explain in ordinary language what C means.

C0 Define variables for “price of one ticket” and “number of tickets sold,” then use them to write a compact expression for total money collected.

C0 If Q = number of sandwiches sold, explain why $Q = 3$ on one day and $Q = 8$ on another day do not change the meaning of Q .

C0 - Name the number of customers.

$$C = \text{number of customers}.$$

Another letter would also work if it were defined clearly.

C0 - Explain what C means.

C is the variable used as a short name for the quantity “number of customers.”

C0 - Use variables for ticket sales.

One possible choice is

$$P = \text{ticket price}, \quad Q = \text{tickets sold}, \quad R = \text{money collected},$$

so

$$R = P \times Q.$$

C0 - Explain changing values of Q .

Q still means “number of sandwiches sold.” Only its value changes from one day to another.

Checkpoint: Is the variable the quantity itself? No. The variable is the symbol used to represent the quantity.



1.4 Subscripts and Collections (C0)

When several quantities belong to the same family, one main variable can be reused with small labels:

Sales on day 1, day 2, day 3 $\longrightarrow S_1, S_2, S_3$.

The small number is a **subscript**. It tells us which member of the collection we mean.

P_1, P_2, P_3 (product prices), C_1, C_2, C_3 (customer counts).

Do not confuse a subscript with an exponent or multiplication:

x_2 = the x labelled 2, x^2 = x squared, $2x$ = 2 times x .

Questions/Tasks.

C0 If P_1, P_2, P_3 are the prices of products 1, 2, and 3, explain what P_2 means.

C0 Write notation for the number of customers on day 3 using C as the main variable.

C0 Explain the difference between x_2 , x^2 , and $2x$.

C0 - Interpret P_2 .

P tells us the quantity is a price; the subscript 2 identifies product 2. Thus P_2 is the price of product 2.

C0 - Write day 3.

Using C for customer count gives
 C_3 .

C0 - Compare the notation.

x_2 : label, x^2 : square, $2x$: multi

Checkpoint: Why use a subscript? It lets one main symbol name a family of related quantities while the subscript identifies one member.



1.5 Generalizing the Size of a Collection: n (C0)

For a fixed number of members, we can write

S_1, S_2, S_3, S_4, S_5 or $S_1, S_2, S_3, \dots, S_{20}$.

To describe a collection with an unspecified number of members, give the final position a variable name. A common choice is n :

$S_1, S_2, \dots, S_n$.

Here n tells us how many members the collection has, and therefore its final position.

$$n = 5 \Rightarrow S_1, S_2, \dots, S_5, \quad n = 20 \Rightarrow S_1, S_2, \dots, S_{20}.$$

This is a **generalization**: one notation works for many possible collection sizes.

Questions/Tasks.

C0 If a store has 12 products and their prices are written $P_1, P_2, \dots, P_n$, what is n ?

C0 If $n = 8$, rewrite $C_1, C_2, \dots, C_n$ with a numerical final subscript.

C0 Explain why $S_1, S_2, \dots, S_n$ is more general than $S_1, S_2, \dots, S_{20}$.

C0 - Find n for 12 products.

There are 12 members, so

$$n = 12.$$

C0 - Replace n by 8.

$$C_1, C_2, \dots, C_n \longrightarrow C_1, C_2, \dots, C_8.$$

C0 - Explain why n is general.

$S_1, S_2, \dots, S_{20}$ fixes the final position at 20. In $S_1, S_2, \dots, S_n$, the value of n can change with the situation.

Checkpoint: What does n do? n gives a general name to the collection size and final position.



1.6 Generalizing a Position: i (C0)

In

$$S_1, S_2, S_3, \dots, S_n,$$

n tells us how far the collection goes. To name one position without choosing it in advance, use an **index**, commonly i :

$$S_i.$$

The value of i tells us which position is being used:

$$i = 1 \Rightarrow S_i = S_1, \quad i = 4 \Rightarrow S_i = S_4, \quad i = 12 \Rightarrow S_i = S_{12}.$$

Thus n gives the collection size, while i selects one position inside it. Compactly,

$$1 \leq i \leq n.$$

For example,

$$P_i = \text{price of product } i, \quad S_i = \text{sales on day } i.$$

Questions/Tasks.

C0 Translate P_3 into ordinary language if P_i means the price of product i .

C0 If $i = 7$, rewrite P_i using a numerical subscript.

C0 If there are 12 products, state the value of n . Can $i = 5$? Can $i = 13$? Explain.

C0 Explain the difference between P_n and P_i .

C0 - Interpret P_3 .

P_3 means “the price of product 3.”

C0 - Replace i by 7.

$$P_i = P_7.$$

C0 - Compare n and i .

If there are 12 products, $n = 12$. Then $i = 5$ is allowed, but $i = 13$ is beyond the final position.

C0 - Distinguish P_n and P_i .

P_n is the price at the final position. P_i is the price at whichever position i currently represents.

Checkpoint: Do n and i do the same job? No. n gives the size or final position; i selects one position within the collection.



1.7 Using Two Indices: i and j (C0)

Sometimes one label is not enough. In a table, one index can identify the row and another the column:

	$j = 1$	$j = 2$	$j = 3$
$i = 1$	a_{11}	a_{12}	a_{13}
$i = 2$	a_{21}	a_{22}	a_{23}
$i = 3$	a_{31}	a_{32}	a_{33}

A general entry is

$$a_{ij},$$

where i identifies the row and j identifies the column. For example,

$$a_{12} = \text{row 1, column 2}, \quad a_{31} = \text{row 3, column 1}.$$

The letters i and j are common choices, but their roles come from how we define them.

Questions/Tasks.

C0 In the table above, locate a_{23} . State its row and column.

C0 Write the notation for the entry in row 3, column 2.

C0 If $i = 2$ and $j = 1$, rewrite a_{ij} using numerical subscripts.

C0 Explain a_{ij} in ordinary language.

C0 - Locate a_{23} .

The first index gives row 2 and the second gives column 3.

C0 - Write row 3, column 2.

$$a_{32}.$$

C0 - Replace i and j .

If $i = 2$ and $j = 1$, then

$$a_{ij} = a_{21}.$$

C0 - Explain a_{ij} .

a_{ij} is the entry identified by position i in one category and position j in another; in a table, row i and column j .

Checkpoint: Can i and j change separately? Yes. For example, a_{12} , a_{13} , and a_{22} identify different entries.



1.8 Combining Indices and Ellipses (C0)

Now the ideas can be combined:

$$x_1, x_2, \dots, x_n.$$

Here x names the kind of quantity, the subscripts identify positions, $\dots$ omits predictable middle members, and n gives the final position.

With two indices,

$$a_{11}, a_{12}, \dots, a_{1n}$$

keeps the first index fixed at 1 while the second index changes. A general member of that row can be written

$$a_{1j}.$$

Questions/Tasks.

- C0 Translate $x_1, x_2, \dots, x_n$ into ordinary language.
- C0 Translate “the first a in row 2, the second a in row 2, and so on, up to column n ” into indexed notation.
- C0 Explain what changes and what stays fixed in $a_{11}, a_{12}, \dots, a_{1n}$.

C0 - Read $x_1, x_2, \dots, x_n$. “The first x, the second x, and so on, up to the n-th x.”	C0 - Write row 2. $a_{21}, a_{22}, \dots, a_{2n}.$	C0 - Explain the indices. In $a_{11}, a_{12}, \dots, a_{1n}$, the first index stays 1 and the second changes until n.
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Checkpoint: Why use compact notation? It saves space while keeping enough information to identify the quantities and positions being discussed.



2 Conceptual Checkpoints and Summary

Problems:
2.1. Checkpoint Review
2.2. From Concrete to Abstract Notation

2.1 Checkpoint Review

Check these ideas:

1. A variable is a symbolic name for a quantity; its meaning comes from its definition.
2. A subscript distinguishes members of one collection.
3. n gives the collection size or final position.
4. i selects one position; i and j can select two positions or categories.
5. ... omits entries only when the intended pattern is clear.

Questions/Tasks.

C0 If P = price of one sandwich, does the letter P automatically mean price? Explain.

C0 If a collection has 10 members, what is n ? If $i = 4$, what does the index i identify?

C0 If $i = 3$ and $j = 2$, what does a_{ij} become?

C0 Explain the difference between the jobs of n and i in $x_1, x_2, \dots, x_n$ and x_i .

C0 - Explain P .

No. The definition P = price of one sandwich gives P that meaning here.

C0 - Distinguish n and i .

$n = 10$. If $i = 4$, then i identifies the fourth position.

C0 - Replace i and j .

$$a_{ij} = a_{32}.$$

C0 - Explain n and i .

n gives the final position or collection size; i identifies one position within that collection.



2.2 From Concrete to Abstract Notation

The progression is

quantity $\longrightarrow$ variable $\longrightarrow$ indexed variables $\longrightarrow$ general size $n \longrightarrow$ general position $i \longrightarrow$ two indices i, j .

For example,

price of a product $\longrightarrow P \longrightarrow P_1, P_2, P_3 \longrightarrow P_1, P_2, \dots, P_n \longrightarrow P_i$.

If a second category is needed, use notation such as

$$a_{ij}.$$

The letters may change later, such as in A_i and S_j , but the same ideas of defining symbols and using indices remain.

